# Related Literature, Citation, and Referencing

A practical lecture on researching sources, writing a Review of Related Literature (RRL), using APA 7th Edition, and incorporating literature into Topic Abstracts and manuscripts

# Learning objectives

**By the end of the lecture, students should be able to do the following:**

- Search for credible and recent research literature.
- Organize sources using Zotero or another citation management tool.
- Extract useful notes, snippets, and study limitations from each source.
- Write related literature through synthesis, not just summaries.
- Apply APA 7 in-text citations and reference list formatting.

# Part 1: Research before writing

Prepare the tools, sources, and notes before drafting the Topic Abstract or manuscript.

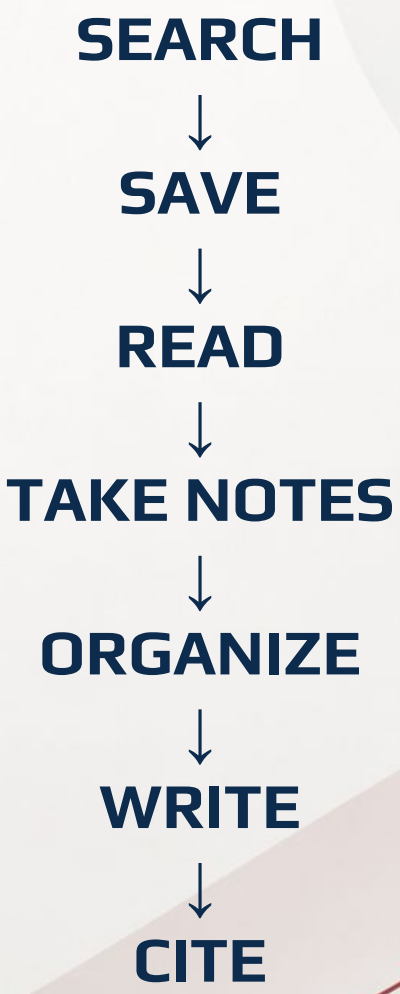
- Do not write first; search for citations later.
- Let the literature shape the problem, gap, and solution.

# Research workflow before writing

**Students should organize the process before drafting paragraphs.**

Do not start by opening Word and writing random paragraphs. Start by building a source library and a clear evidence trail for every important claim.

**Research first. Write after you understand the sources.**



# Step 1: Install and set up Zotero

- Install **Zotero Desktop** as your main reference library.
- Install the **Zotero Browser Connector** to save article details from databases and journals.
- Enable **Zotero Word** or **Google Docs** integration for inserting citations while writing.
- Set the **citation style** to **APA 7th Edition**.
- Check imported details manually because Zotero is helpful but not always perfect.

**Zotero helps you organize references, but you are still responsible for accuracy.**

# Step 2: Organize your source library

## Create collections

- **Main folder:** Thesis or Capstone title
- **Subfolders:** problem, related systems, algorithms, methodology, evaluation, research gap
- Attach **PDFs** directly to the Zotero entry
- Use **tags** such as Problem, Method, Dataset, Limitation, Gap, and POC

## Check every entry

- Author names
- Publication year
- Article title and journal title
- Volume, issue, pages, and DOI
- Correct APA 7 citation output

# Step 3: Begin with the research problem

- Do **not** search using only the full thesis title.
- First, identify the **exact problem** you want to solve.
- Break the problem into **searchable concepts** and **keywords**.
- Example concepts: **customer churn, traffic prediction, stockout, text classification, malicious URLs.**
- Use several **keyword combinations** instead of only one search phrase.

**Clear problem first → better keywords → more relevant literature.**

# Step 4: Search for refereed journals

- Prioritize **peer-reviewed** or **refereed journal articles** and **conference papers**.
- Useful sources: **IEEE Xplore, ACM Digital Library, ScienceDirect, Springer, Wiley, Scopus-indexed journals, and Web of Science-indexed journals**.
- Use **Google Scholar** as a search tool, not as the publisher.
- Prioritize **recent sources** from the last **five** years, unless the source is foundational.
- For a 2026 study, recent sources are generally **2022–2026**.

# Step 5: Use stronger search strategies

## Useful operators

• **AND** narrows the search:

Example: machine learning **AND** inventory forecasting

• **OR** finds similar terms

Example : inventory forecasting **OR** demand forecasting

• **Quotation marks** search an **exact** phrase

Example : **“**demand forecasting**”**

• Combine terms

Example : **“**demand forecasting**”** **AND** **“**machine learning**”** **AND** retail

## Suggested keyword sources

• **Words** from your research problem

• **Terms** used in good articles you already found

• **Algorithm** or **method** names

• **Dataset** or **domain** terms

• **Limitations** and **future work** from related studies

# Step 6: Check article relevance

- Do not download a paper only because its title contains AI, system, analytics, or machine learning.
- Ask what **problem the paper solved**, what **data** it used, and what **method** it applied.
- Check the **findings, limitations**, and **connection** to your proposed study.
- A useful source should **support the problem**, **method**, **algorithm**, **dataset**, **evaluation**, or **gap**.
- If you cannot explain why you need the source, do not force it into the manuscript.

# Step 7: Download and save the actual article

- Whenever possible, download the **full PDF**, not only the abstract or preview.
- Save the article in Zotero together with its complete publication details.
- **Do not** cite sources **generated by AI** unless you personally verify the actual article.
- If you cannot locate and read the real source, do not include it in your manuscript.
- Keep the DOI or official publisher link for checking later.

**No actual article, no citation.**

# Step 8: Read efficiently

- Start with the **title** and **abstract** to confirm relevance.
- Read the **introduction** to understand the **problem and purpose**.
- Check the **methodology** for the **algorithm, model, dataset, or procedure**.
- Read the **results** and discussion to understand what was found.
- Check the **conclusion**, **limitations**, and **future work** for possible research gaps.

# How to jot down snippets from each source

**A snippet is a useful idea with context—not a sentence to copy-paste.**

- Save the **original idea** and the **page** or **section** where it came from.
- Write your **own** understanding in simple words.
- Record why you need it: **problem, method, algorithm, gap, or evaluation**.
- Draft a **possible paraphrase** and **add the correct citation immediately**.
- **Avoid** collecting random highlighted sentences with **no purpose**.

**Always ask: "Why am I saving this information?"**

# Create a Literature Matrix

**This helps students move from collecting articles to comparing and synthesizing them.**

**Recommended columns**

| Author/Year | Problem | Method | Data | Major Finding | Limitation | Relevance to Study |
|---|---|---|---|---|---|---|
| Santos et al. (2024) | Demand forecasting | XGBoost | Historical sales data | Improved demand prediction | Limited to historical sales patterns | Supports the forecasting component |
| Cruz & Lim (2025) | Stockout prediction | Random Forest | Inventory and sales data | Identified products at risk of stockout | Did not include automated recommendations | Supports stockout analysis and risk indicators |
| Reyes et al. (2023) | Inventory monitoring | Monitoring system | Inventory records | Improved tracking of stock levels | Focused only on monitoring | Shows the gap in monitoring-only systems |

**Tip: The matrix shows patterns, common limitations, and possible research gaps.**

# Part 2: Writing Related Literature

After research and notes, students can start building the actual discussion.

- Related literature explains the problem and supports the proposed direction.
- It must lead to the gap and objectives.

# What Related Literature should do

- Establish that the research problem **really exists**.
- Explain why the problem is **important**.
- Describe **existing technologies**, **methods**, or **systems**.
- Identify **limitations in previous studies**.
- Support the **proposed solution, computational method, and proof of concept**.

# From Summaries to Synthesis

## Weak approach

- One paragraph per article
- Only states what each author did
- No comparison among studies
- No clear connection to the gap
- Literature feels like a list

## Stronger approach

- Groups studies by **theme**
- **Compares** methods, data, findings, and limitations
- Explains what is **common** and what is **missing**
- Leads to the **proposed study**
- Shows the student **understood the field**

# A simple formula for RRL paragraphs

- **IDEA**: introduce the **issue** or **theme**.
- **EVIDENCE**: present what **several studies** say.
- **COMPARISON**: explain **similarities** or **differences** among studies.
- **LIMITATION**: identify what still **appears missing** or **unresolved**.
- **CONNECTION**: explain how this **supports** your proposed study.

**Do not just report studies. Connect them.**

# Organize RRL by themes, not by Author

- Theme 1: Inventory management problems, stockout, overstocking, inaccurate records.
- Theme 2: Demand forecasting and historical sales data.
- Theme 3: Machine learning approaches such as regression, Random Forest, XGBoost, or neural networks.
- Theme 4: Decision support, recommendations, and inventory optimization.
- Theme 5: Limitations and research gap.

# Problem → Literature → Gap → Solution

- **PROBLEM**: What is happening?
- **EVIDENCE**: What research proves or explains the problem?
- **EXISTING WORK**: What has already been done?
- **LIMITATION / GAP**: What remains unaddressed?
- **PROPOSED SOLUTION**: How will your study respond to the gap?

# Part 3: Using literature in the Topic Abstract

The Topic Abstract gives the short version of the research argument.

- Do not include a full RRL.
- Support key claims with citations.

# Topic Abstract Structure

- **Problem:** State the specific issue being addressed.
- **Evidence:** Use recent literature to support the problem.
- **Existing approach or limitation**: Show what current studies/systems already do and what is missing.
- **Proposed solution:** Briefly introduce the system or study.
- **Objectives or expected output**: State what will be developed, evaluated, or produced.

# Where citations are needed in the Topic Abstract

## Needs citation

- Statistics or factual claims
- Claims about existing problems
- Claims about technologies or algorithms
- Statements about previous studies
- Claims about limitations or gaps

## May not need citation

- Your proposed system features
- Your intended users
- Your own objectives
- Your proposed output
- Your research questions

# Example: weak vs. improved abstract writing

- **Weak:** “Cybersecurity threats have increased. Employees lack awareness.” According to whom?
- **Improved**: Cite literature showing that phishing and low awareness remain problems.
- Then identify what existing training or systems lack.
- Then present the proposed adaptive cybersecurity awareness platform.
- The literature should justify the problem and the proposed direction.

# Example: weak vs. improved abstract writing

## Improved:

Phishing continues to be a major cybersecurity concern because employees may still respond to deceptive emails even after awareness training (Yeoh et al., 2022; Canfield et al., 2023). Research also shows that the effectiveness of training may decrease over time and that repeated practice and feedback are important for improving phishing awareness (Sikolia et al., 2023; Canfield et al., 2023).

These findings can then be used to identify the **limitation of existing training** and justify the proposed **adaptive cybersecurity awareness platform**.

**Claim → Citation → Limitation from literature → Research gap → Proposed solution**.

# Part 4: Expanding into the full manuscript

The manuscript expands the short argument from the Topic Abstract into detailed discussion.

- Background establishes the problem.
- RRL analyzes studies by theme.
- Synthesis leads to the research gap.

# How literature appears in the manuscript

- Background of the Study: introduce the problem, context, importance, and need for the study.
- Review of Related Literature: discuss themes, methods, technologies, findings, and limitations.
- Synthesis: explain what the reviewed studies collectively show.
- Research Gap: state what remains unresolved based on the reviewed literature.
- Objectives: align objectives with the established problem and gap.

# Alignment to check

- Title should match the problem and proposed solution.
- Problem should be supported by recent and credible literature.
- Objectives should directly address the stated problem.
- Method or algorithm should be justified by literature, not chosen randomly.
- Proof of Concept should show that the proposed computational solution can work.

# Part 5: APA 7  Citation and References

- Citation = short source identification in the text.
- Reference = complete source details at the end.

# APA 7: in-text citation basics

Use citations where the idea or evidence appears.

## Parenthetical citation

Machine learning has been applied to demand forecasting (Santos & Reyes, 2024).

## Narrative citation

Santos and Reyes (2024) applied machine learning to demand forecasting.

## Remember

- One author: (Santos, 2024)
- Two authors: (Santos & Reyes, 2024)
- Three or more: (Santos et al., 2024)
- Inside parentheses use “&”; in a sentence use “and”.

# APA 7: reference list essentials

**Citation is the short ID; reference is the complete source information.**

- Every in-text citation must appear in the References.
- Every item in the References should be cited in the manuscript.
- Arrange entries alphabetically by first author surname.
- Use hanging indent, double spacing, DOI when available, and APA capitalization.

## Journal article format

*Author, A. A., & Author, B. B. (Year). Title of article. Title of Journal, volume(issue), pages. DOI*

# APA 7 reference examples

- Journal article: Author, A. A., & Author, B. B. (Year). Title of article. Title of Journal, volume(issue), pages. DOI
- Conference paper: Author, A. A. (Year). Title of paper. In Proceedings of the Conference (pp. xx–xx). Publisher. DOI/URL
- Use sentence case for article titles and title case for journal names.

# Part 6: Tips, suggestions, and common issues

- The strongest papers show clear evidence, alignment, and synthesis.
- The weakest papers only collect citations.

# Common mistakes to avoid

- **Unsupported claims**: “many users,” “most effective,” “best algorithm,” or “highly accurate” without evidence.
- **Citation dumping**: many citations placed at the end without explaining what they support.
- **Old** or **unrelated sources** used only to meet the required number of references.
- **Fake, unverifiable, or AI-generated references**.
- **Research gap** based only on **assumptions** instead of the reviewed literature.

# Practical tips for students

- Use a literature matrix from the start.
- Write notes in your own words before drafting the manuscript.
- Use several sources to support major claims, not just one article for everything.
- Check the limitations and future work sections of good papers.
- Do not decide the algorithm first; understand the problem and literature first.

# Class activity

- Each group must bring 5 recent refereed articles saved in Zotero.
- Prepare a literature matrix with Author/Year, Problem, Method, Findings, Limitation, and Relevance.
- Identify which article supports the problem, method, technology, gap, and proof of concept.
- Write one synthesized paragraph using at least 3 sources.
- Check if the citations and reference entries match APA 7.

# Final reminder

**Research first. Organize sources. Read the actual articles. Take notes. Synthesize. Identify the gap. Then write.**

Problem → Evidence → Existing Work → Gap → Proposed Solution

This is the real purpose of related literature.